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الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

MST129-MT129
Applied Calculus

Final Examination
Summer Semester 2025/2026

Time Allowed: 2 Hours

Instructions:

- Total Marks: **50**
- This exam consists of 5 questions
- ALL questions must be answered in the External Answer booklet
- Be sure you write your **name and ID** on the External Answer booklet
- **Calculators** are allowed

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Q-1: [5×2 marks] Choose the correct answer:

1.1: The solution of $3lnx - ln3x = 0$ *is*:

a) 0
b) $\sqrt{3}$
c) 3
d) None of the above

1.2: All solutions of the differential equation $y' = 5y$ *and* $f(0) = 3$ *is*:

a) $y=5e^3$
b) $y=3e^{5x}$
c) $y= 5e^{1.5x}$
d) None of the above

1.3: $\int_1^2 5\,dx =$

a) 5
b) 3
c) 1
d) None of the above

1.4: The value of $\int x.x^2\,dx$ is:

a) $\frac{x^4}{4}+c$
b) $\frac{x^3}{3}+c$
c) $\frac{x^2}{2}+c$
d) none of the above

1.5: If $y = x^5 tanx$, then $\frac{dy}{dx} =$

a) $5x^4sec^2(x)+x^5sec^2(x)$
b) $5x^4\tan(x) + x^5sec^2(x)$
c) $5x^4\tan(x)$
d) None of the above

Each question is worth 10 marks. Answer the following questions:

Q-2:

a) **[6 marks]** Let $f(x) = \frac{1}{3}x^3 - 2x^2 + 3x + 1 \qquad -1 \le x \le 6$

Find the absolute maximum and absolute minimum.

b) **[4marks]** Use implicit differentiation of the equation $x^3y^2 - 4x^2 = -3$ *to* find the equation of the tangent line to the graph at the designated point; $P(1,1)$.

Q-3:

a) **[5 marks]** Find the x-intercept of $y= (x-1)^2 \ln(x+1)$

b) **[5 marks]** use logarithmic differentiation to find the derivative of the function: $y= (\sqrt{x^2+5})e^{x^2}$

Q-4:

a) **[2×3 marks]** Evaluate the following integrals:

i) $\int_{-2}^{2}(e^x + e^{-x})\,dx.$

ii) $\int cot(x) + \cos(3x) + \frac{1}{7x})\,dx$

b) **[4marks]** Use integration by substitution to evaluate $\int \frac{e^x}{1-e^x}\,dx$

Q-5:

a) **[5 marks]** Find the area of the region bounded by the graphs of $y=x^2 + 2x + 3$ *and* $y = 2x + 4$

b) **[5 marks]** Use integration by parts to evaluate $\int x\ (x+5)^5\,dx$

End of questions

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